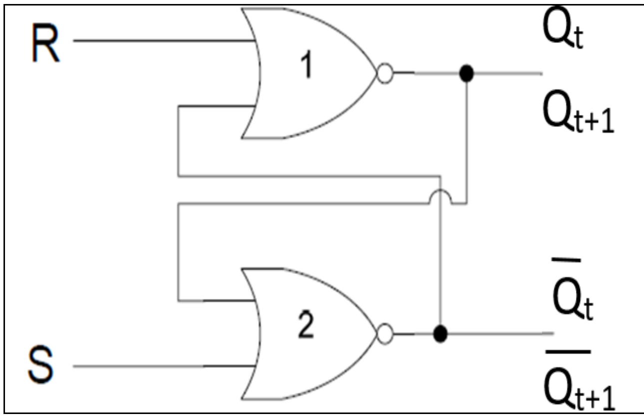
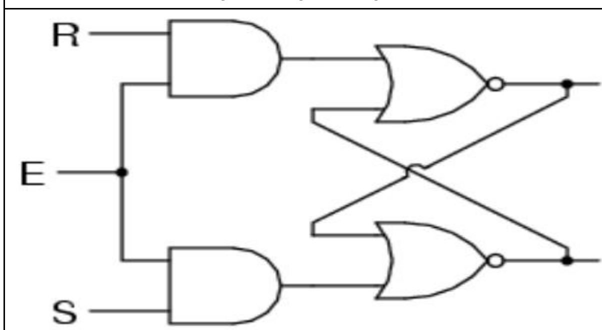
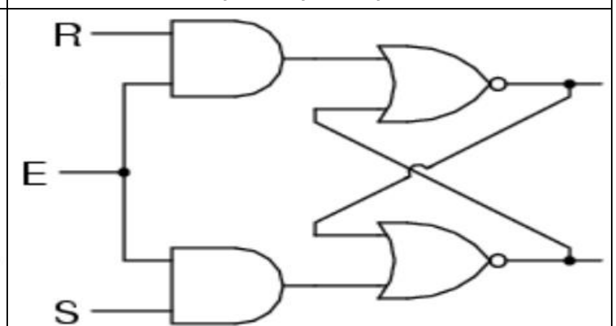
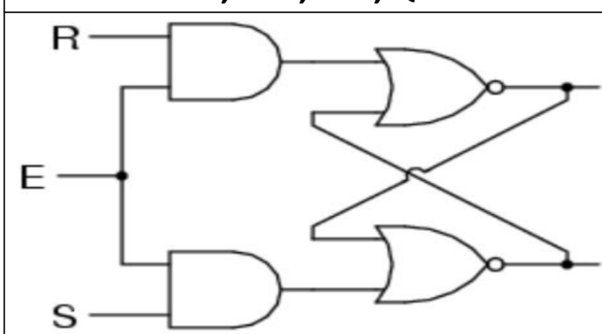
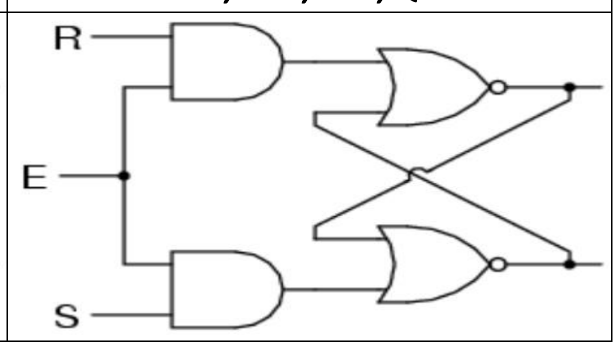
1. *Following is the table for NOR BASED* ***S-R Latch****? Use the table to process all 8 possible cases for the* ***GATED S_R Latch?***

NOTE: *Process each & every gate in all 8 cases*



Input		Output
S	R	Q_{t+1}
0	0	Q_t
0	1	0
1	0	1
1	1	invalid

CASE I	**CASE-II**
S=0,R=0,E=1,Q=0	**S=0,R=0,E=1,Q=1**
	
S=1,R=0,E=1,Q=0	**S=1,R=0,E=1,Q=1**
	

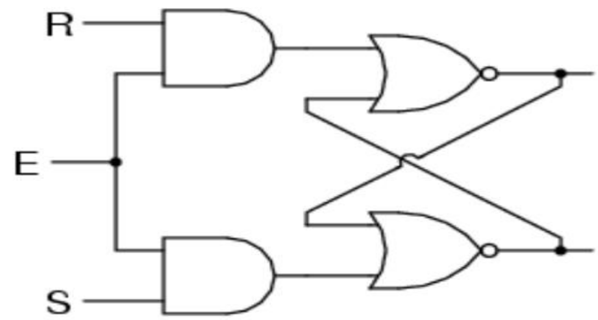
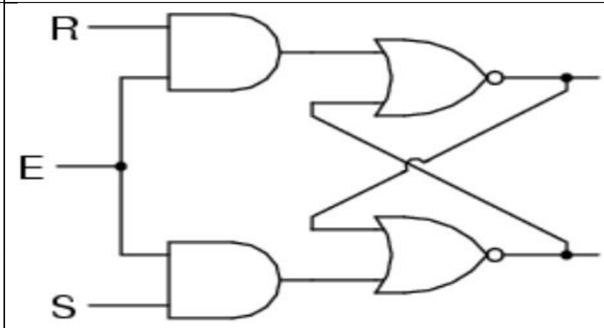
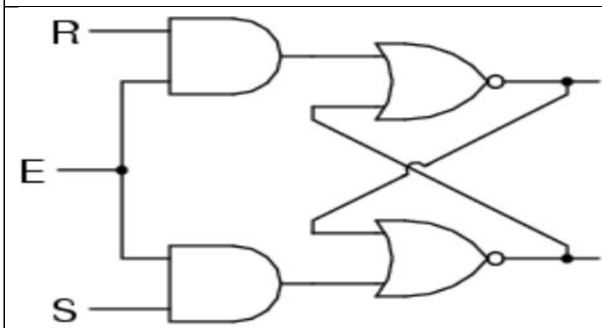
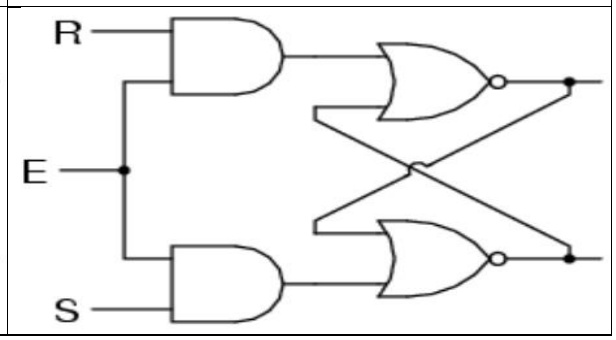
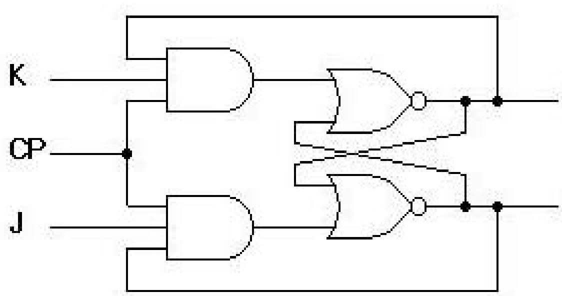
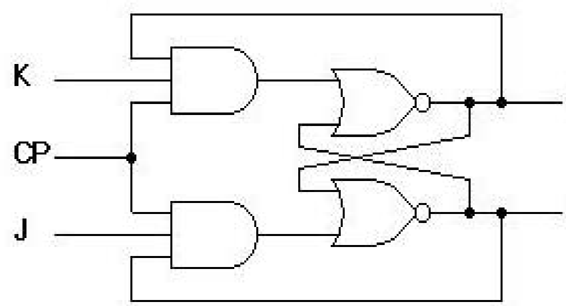
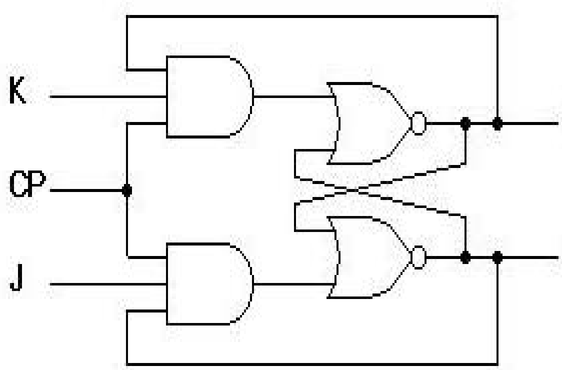
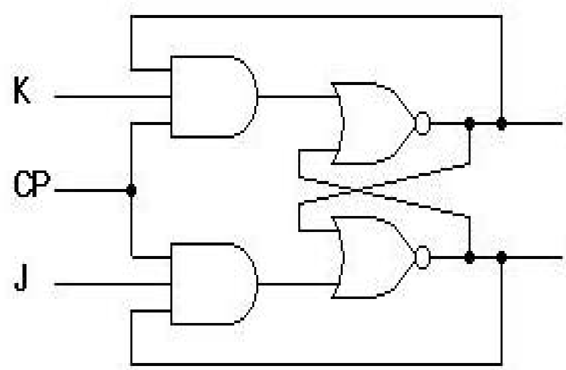
S=0,R=1,E=1,Q=0	**S=0,R=1,E=1,Q=1**
	
S=1,R=1,E=1,Q=0	**S=1,R=1,E=1,Q=1**
	

TABLE:

INPUT			**OUTPUT**
E	**S**	**R**	**Q_{T+1}**
0	X	X	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

*Process the POSITVE CLOCK EDGE **JK FLIP FLOP** circuit given below for all possible cases and fill the table accordingly?*

CIRCUIT:

CLK=0	CLK=1
	
CASE I	**CASE-II**
J=0,K=0,Q=0	**J=0,K=0,Q=1**
	
J=0,K=1,Q=0	**J=0,K=1,Q=1**

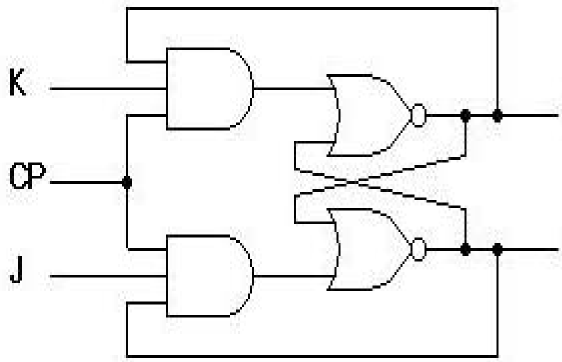
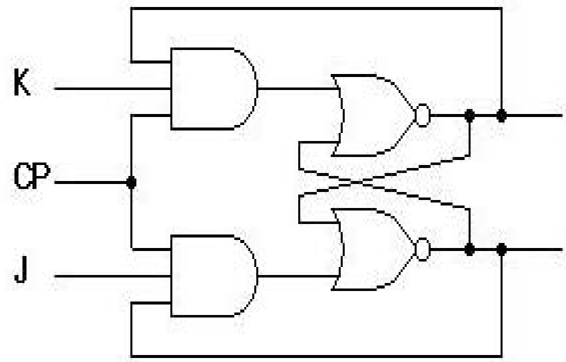
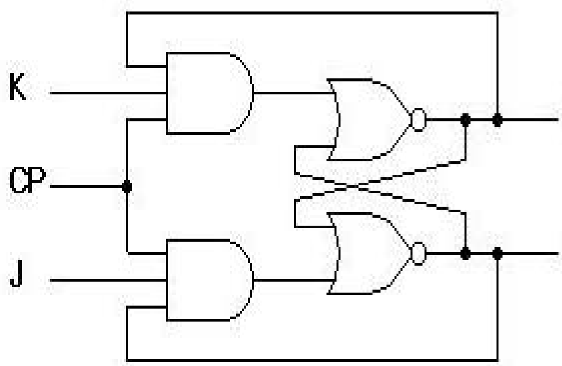
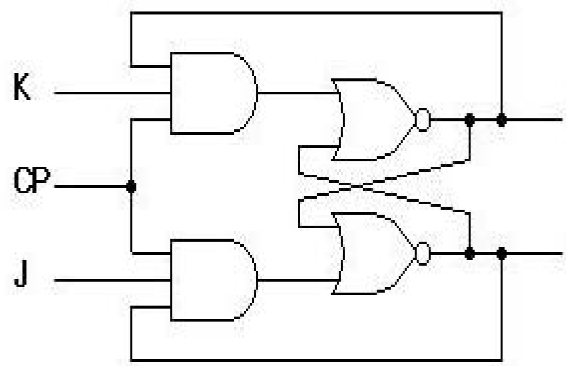
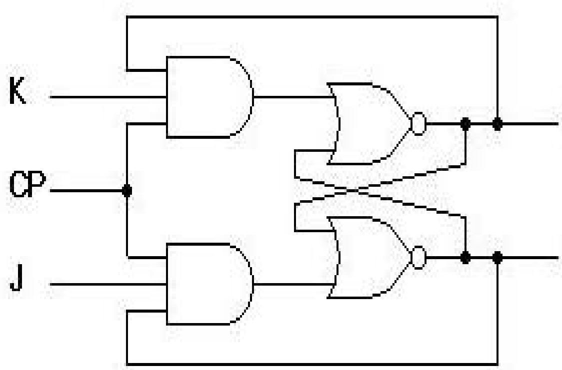
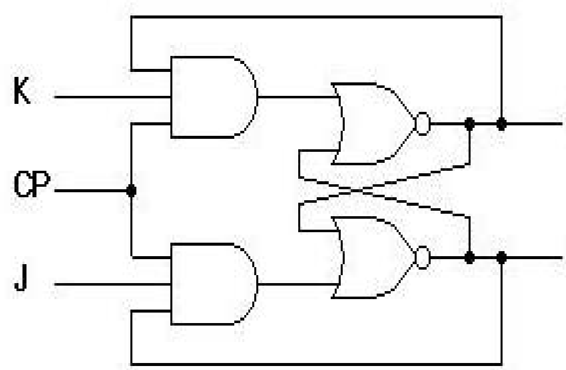
	
J=1,K=0,Q=0	**J=1,K=0,Q=1**
	
J=1,K=1,Q=0	**J=1,K=1,Q=1**
	

TABLE:

CLK	J	K	Q_{t+1}	$\overline{Q_{t+1}}$
0	X	X		
1	X	X		
↑	0	0		
↑	0	1		
↑	1	0		
↑	1	1		

*Modify given circuit to introduce Asynchronous **CLEAR** and **SET** inputs fill table?*

***HINT:** For reference kindly read Section **7-2** of DIGITAL FUNDAMENTAL by Thomas Floyd, 10^{th} Edition*

CIRCUIT:

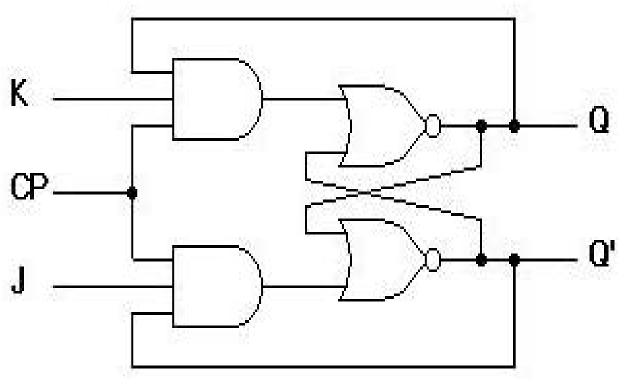
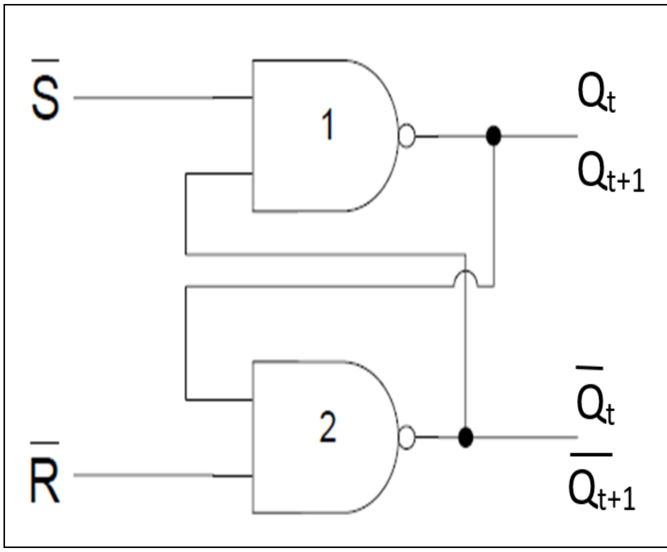


TABLE:

SET	CLEAR	Q_{t+1}	$\bar{Q}_{t+1}$
0	0		
0	1		
1	0		
1	1		

2. *Following is the table for NAND BASED **S-R Latch**? Use the table to process all 8 possible cases for the **GATED S_R Latch?***
   ***NOTE:** Process each & every gate in all 8 cases*

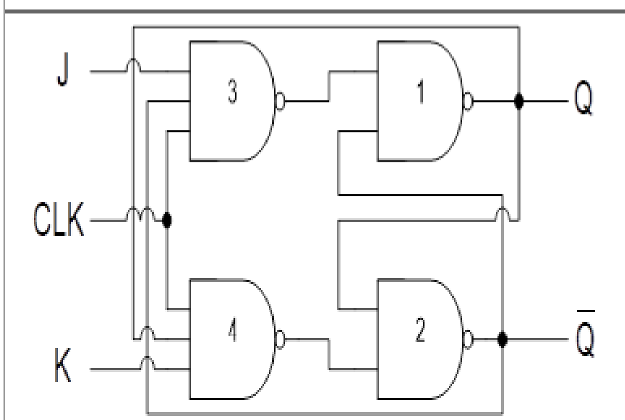
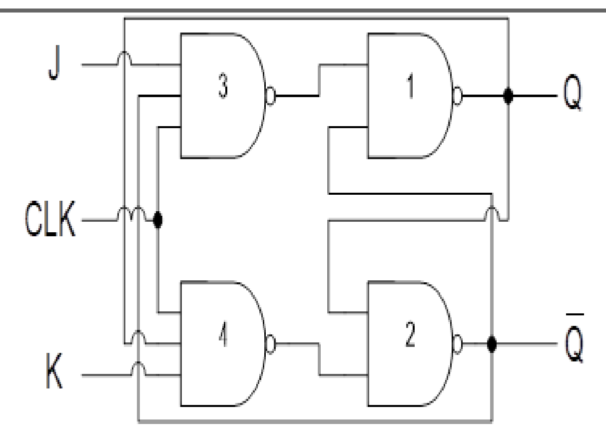


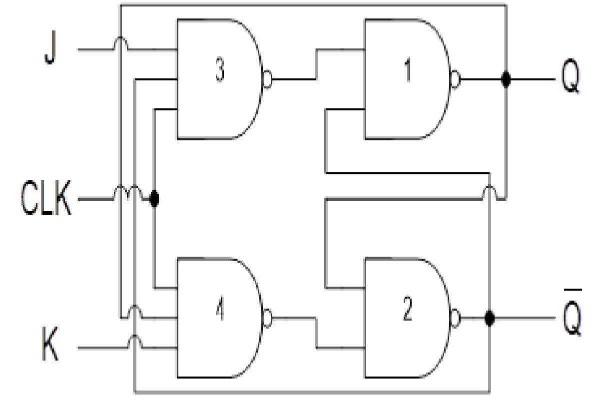
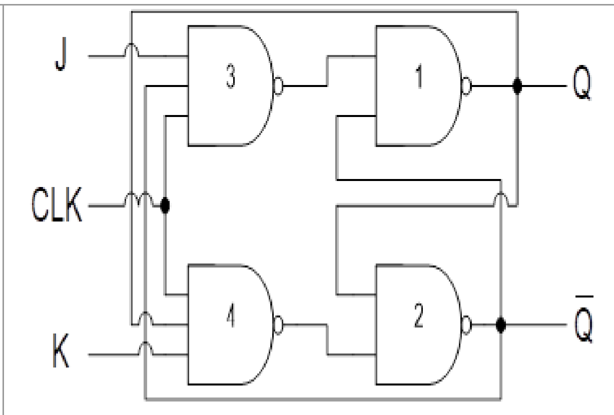
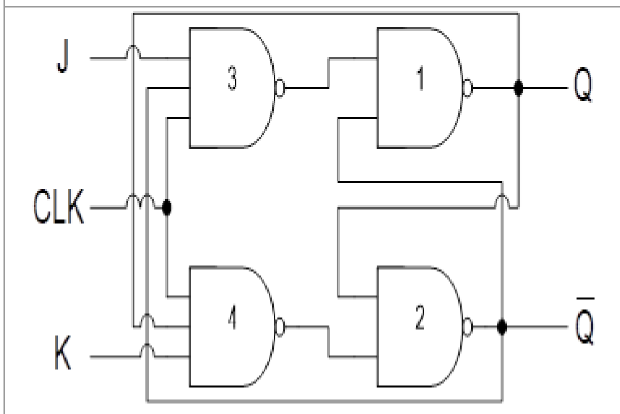
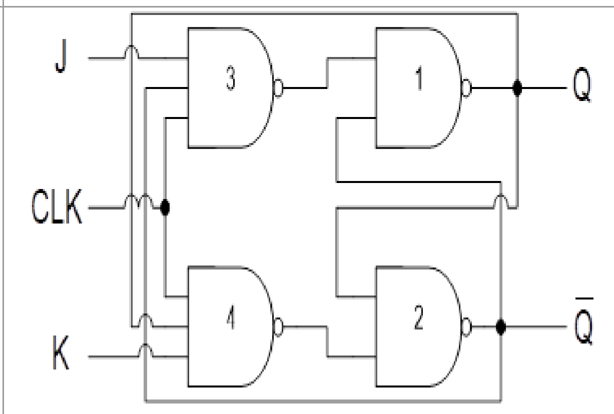
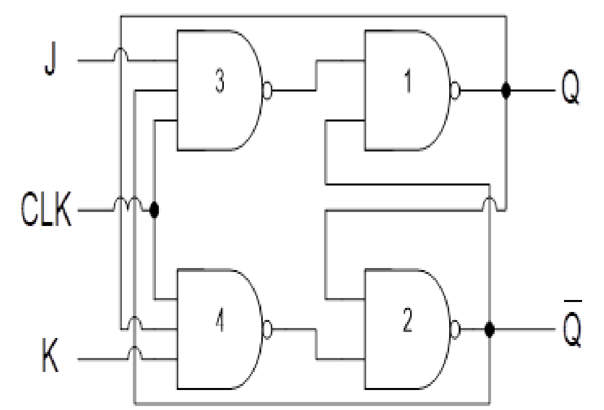
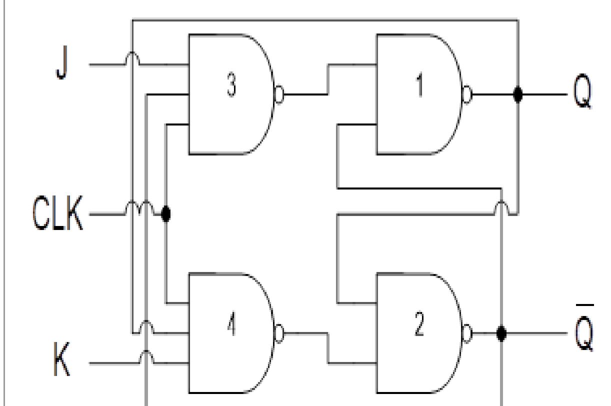
INPUT		OUTPUT	
S	R	Q_{t+1}	$\overline{Q}_{t+1}$
0	0		
0	1		
1	0		
1	1		

- *Process the POSITVE CLOCK EDGE JK FLIP FLOP circuit given below for all possible cases and fill the table accordingly?*

NOTE:** Process each and every block no marks will be given for direct answer. No need to process NAND LATCH use table filled in **Part A

CIRCUIT:

CLK=0	CLK=1
	
CASE I	**CASE-II**
J=0,K=0,Q=0	**J=0,K=0,Q=1**

	
J=0,K=1,Q=0	**J=0,K=1,Q=1**
	
J=1,K=0,Q=0	**J=1,K=0,Q=1**
	
J=1,K=1,Q=0	**J=1,K=1,Q=1**

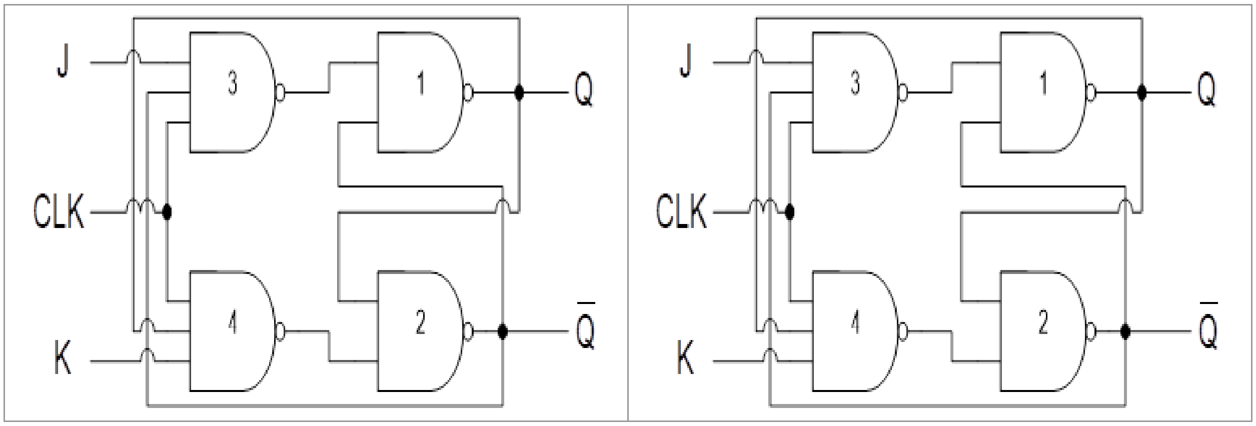


TABLE:

CLK	J	K	Q_{t+1}	$\overline{Q_{t+1}}$
0	X	X		
1	X	X		
↑	0	0		
↑	0	1		
↑	1	0		
↑	1	1		

*Modify given NAND Based JK Flip Flop circuit to introduce Asynchronous **CLEAR** and **SET** inputs fill table?*

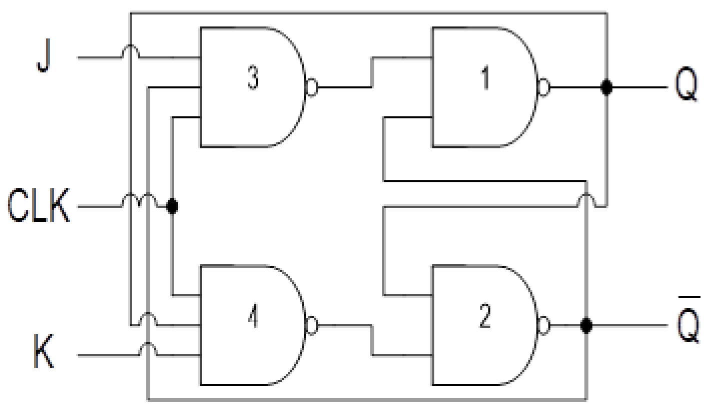
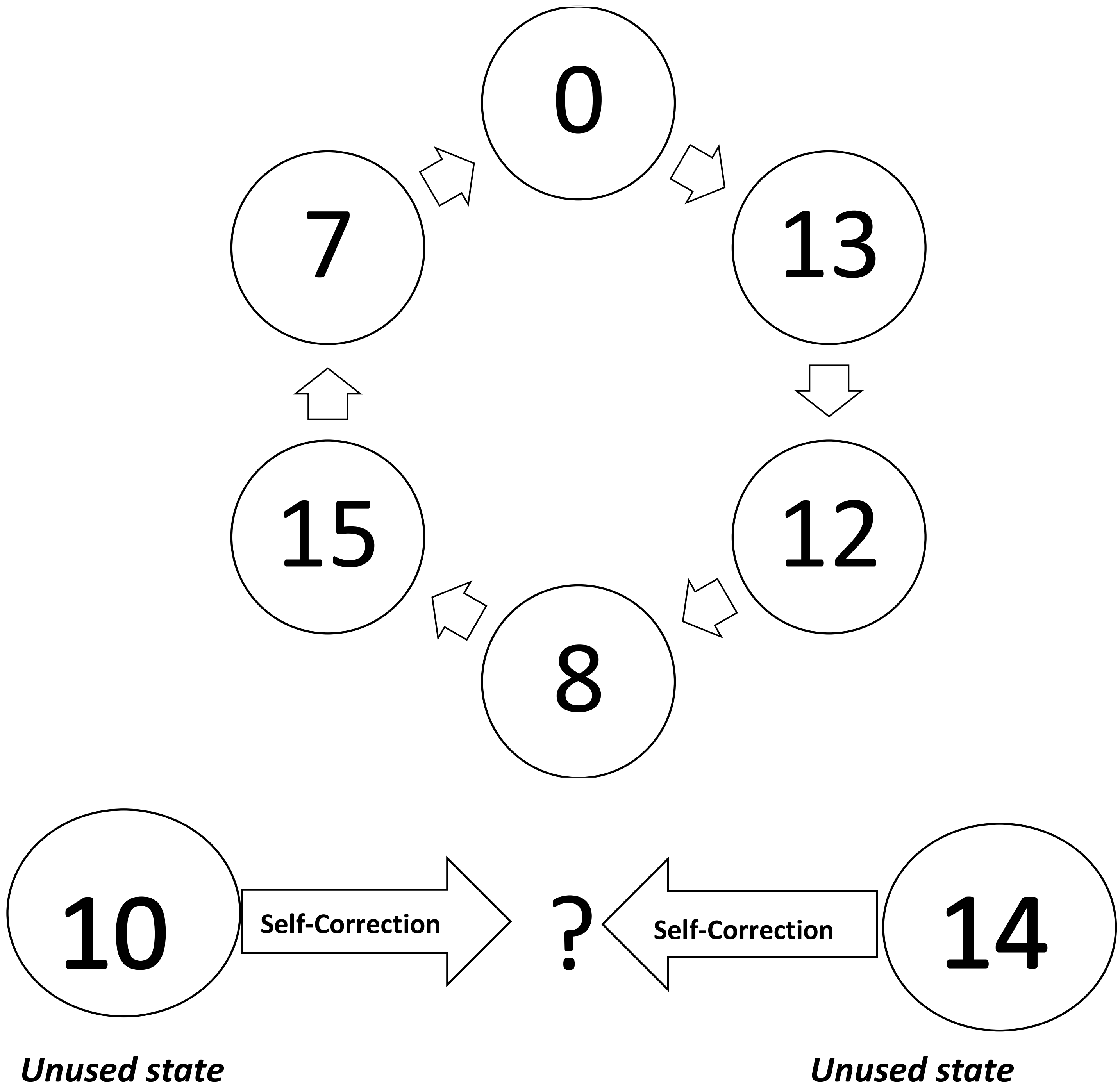


TABLE:

SET	CLEAR	Q_{t+1}	$\overline{Q}_{t+1}$
0	0		
0	1		
1	0		
1	1		

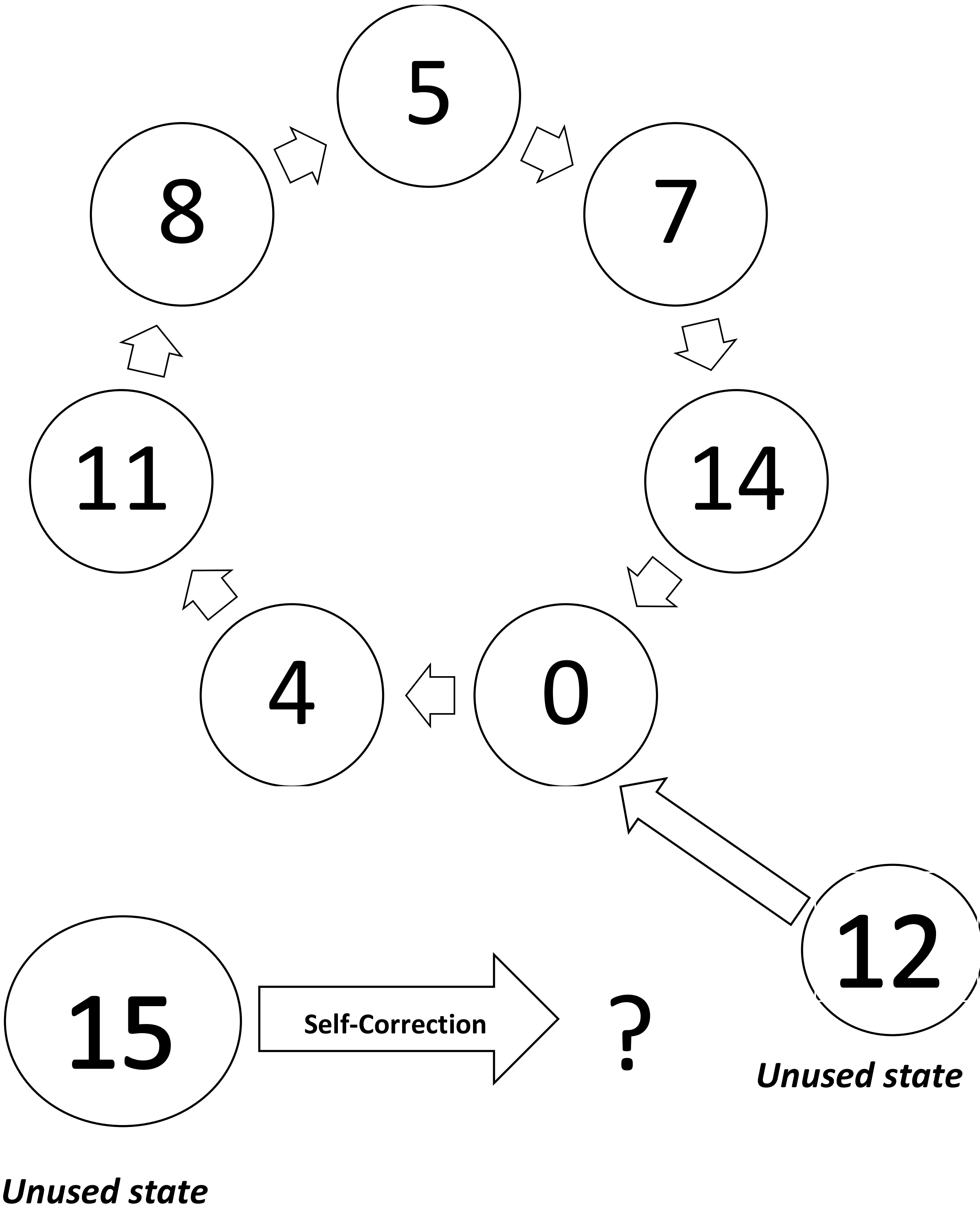
3. *Design the sequential circuit using **T-Flip-flop** specified by the state diagram given below? find self-correcting state for unused state 10, 14?*



State Table

SIMPLIFICATION & EXPRESSIONS

4. *Design the sequential circuit using **JK-Flip-flop** specified by the state diagram given below? find self-correcting state for unused state 15?*



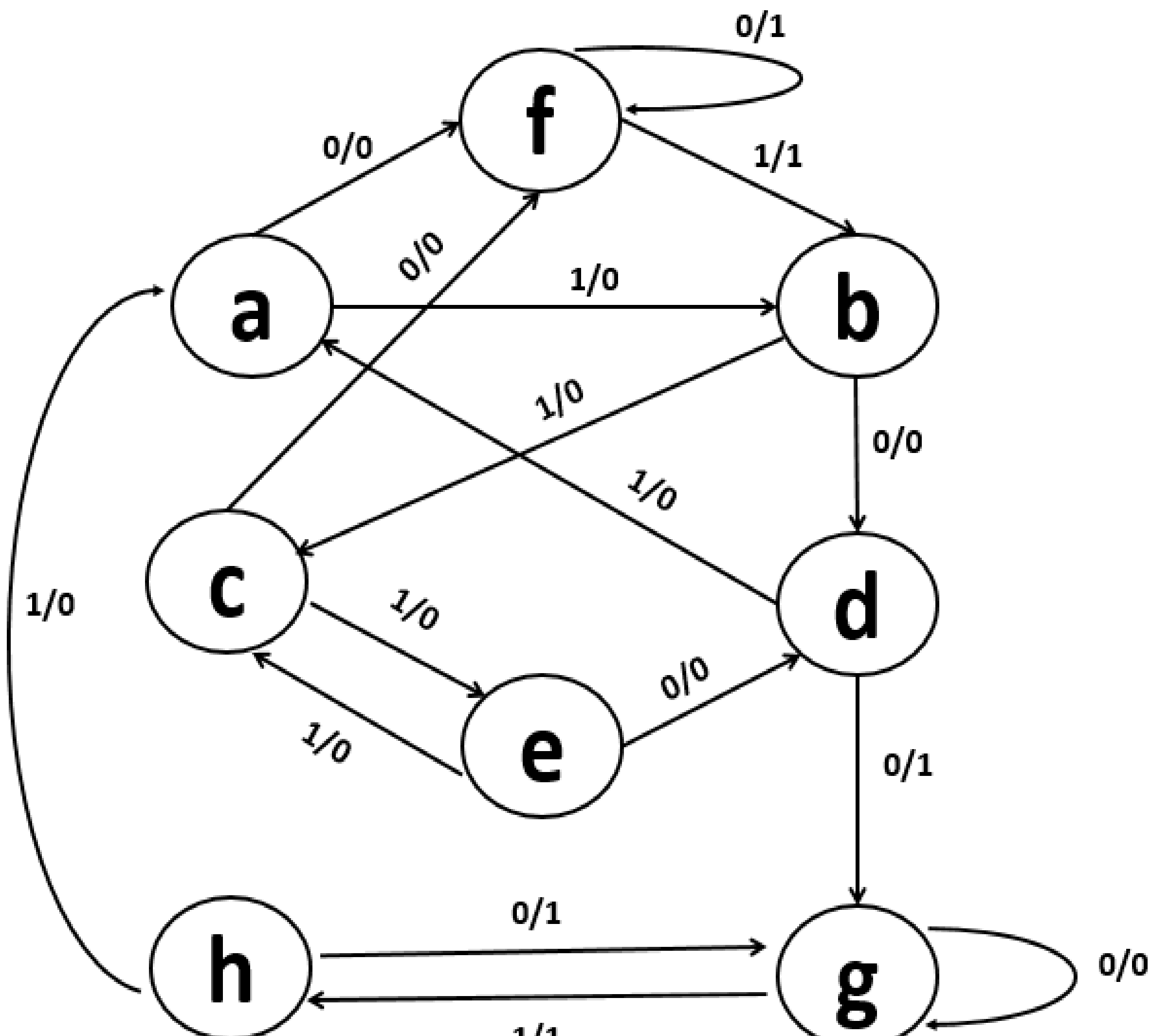
State Table

SIMPLIFICATION & EXPRESSIONS

5. *For* ***STATE DIAGRAM*** *given below determine output where input sequence is 01110010011 where starting state is* ***a***

INPUT	0	1	1	1	0	0	1	0	0	1	1	0	1
STATE	a												
OUTPUT													

<u>State Diagram</u>



*Produce **State Table** for **State Diagram***

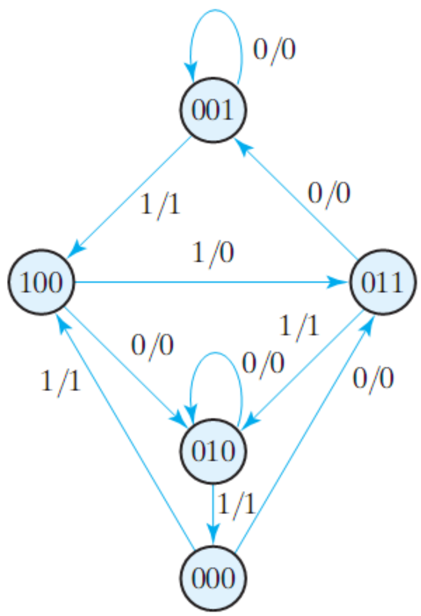
*Generate **Reduced State Table***

*Produce **Reduced State Diagram***

*For **Reduced State Diagram** produced in part d, determine output where input sequence is 01110010011 where starting state is **a***

INPUT	**0**	**1**	**1**	**1**	**0**	**0**	**1**	**0**	**0**	**1**	**1**	**0**	**1**
STATE	**a**												
OUTPUT													

6. *A sequential circuit has three flip-flops A, B, C ; one input 'x' ; and one output 'y'. The state diagram is shown below. The circuit is to be designed by treating the unused states as don't-care conditions. Where 101 is an unused state find self-correction for* ***x=0 & x=1.*** *Use* ***T flip-flops*** *to the circuit design.*



8. *For the given state table perform all of the following.*
    a. *Draw the corresponding state diagram*
    b. *Draw Reduced State Table*
    c. *Draw Reduced State Diagram*

TABLE:

	Next State		Output	
Present State	$x = 0$	$x = 1$	$x = 0$	$x = 1$
a	f	b	0	0
b	d	c	0	0
c	f	e	0	0
d	g	a	1	0
e	d	c	0	0
f	f	b	1	1
g	g	h	0	1
h	g	a	1	0

STATE TRANSITION DIAGRAM:

REDUCED STATE TABLE:

PRESENT STATE	NEXT STATE		OUTPUT	
	X=0	X=1	X=0	X=1

REDUCED STATE DIAGRAM:

For **STATE DIAGRAMS** given below convert your university id to binary and determine **Output** and **NEXT State**

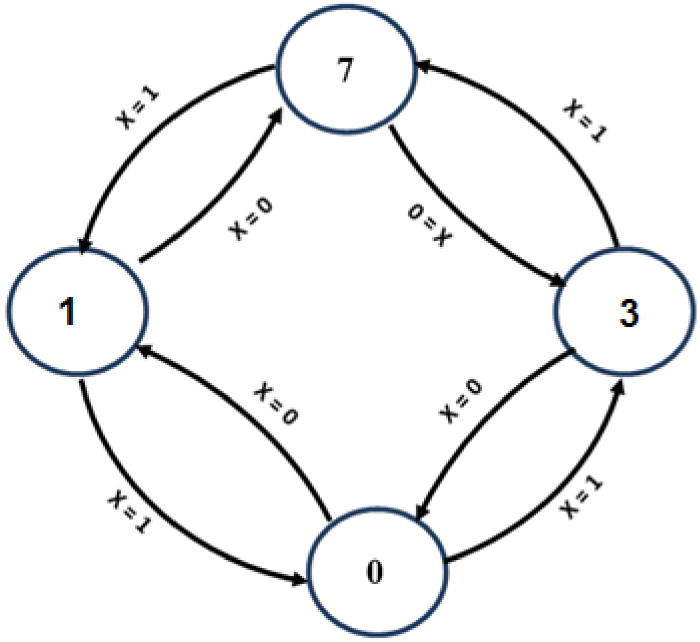
ACTUAL STATE DIAGRAM

INPUT																								
STATE																								
OUTPUT																								

REDUCED STATE DIAGRAM

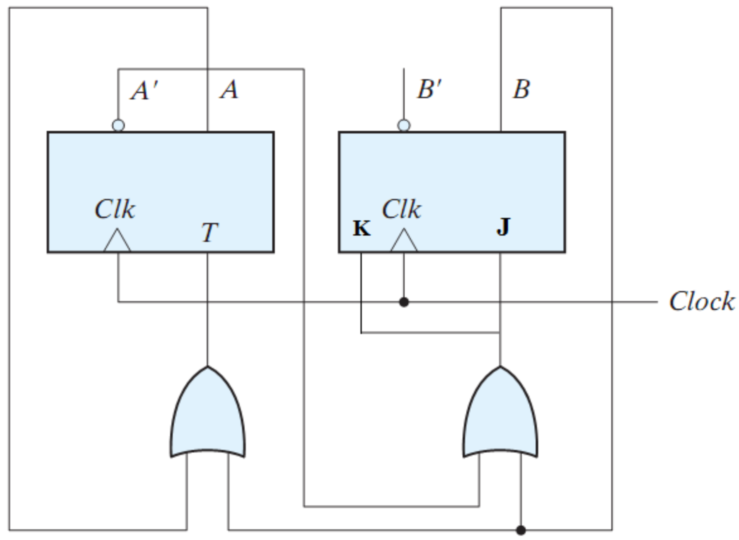
INPUT																								
STATE																								
OUTPUT																								

9. Design circuit for following state diagram using **JK flip Flop**?

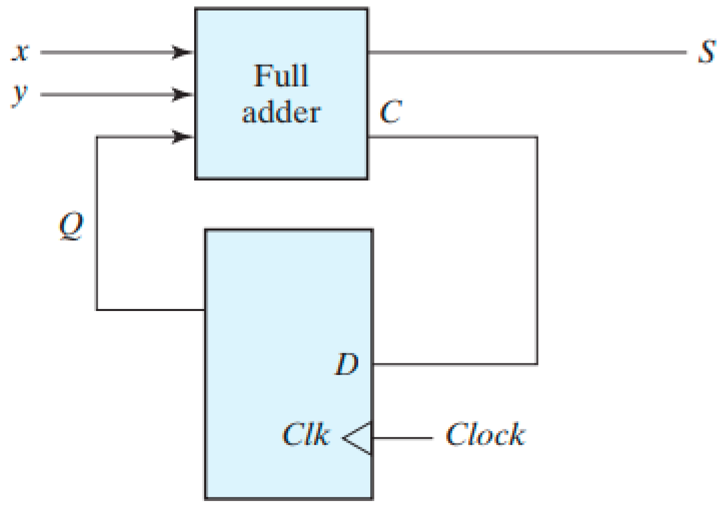


TABLE, SIMPLIFICATION & EXPRESSIONS

10. Derive the state table and the state diagram of the sequential circuit shown in the given figure. Regenerate equivalent circuit using just D Flip Flop

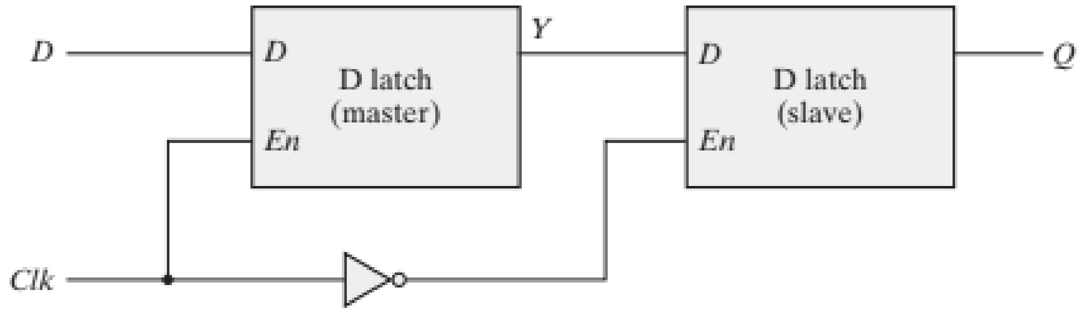


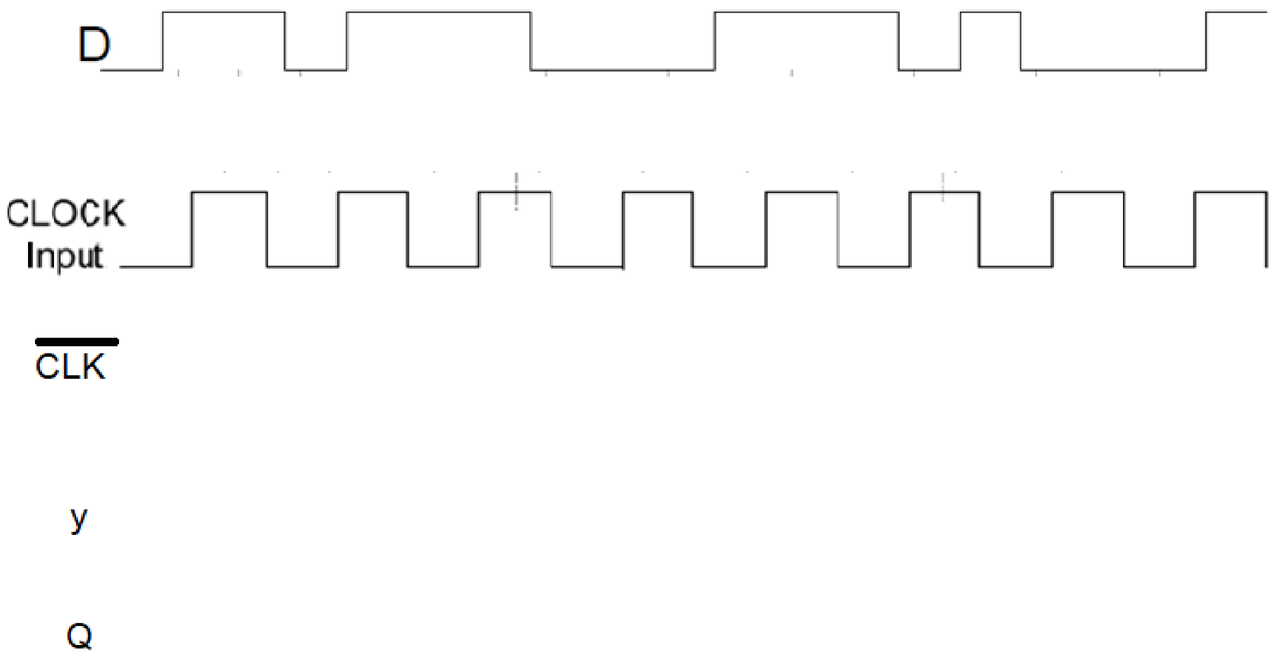
11. *A sequential circuit has one flip-flop Q, two inputs x and y, and one output S . It consists of a full-adder circuit connected to a D flip-flop, as shown in Fig. Derive the state table and state diagram of the sequential circuit.*



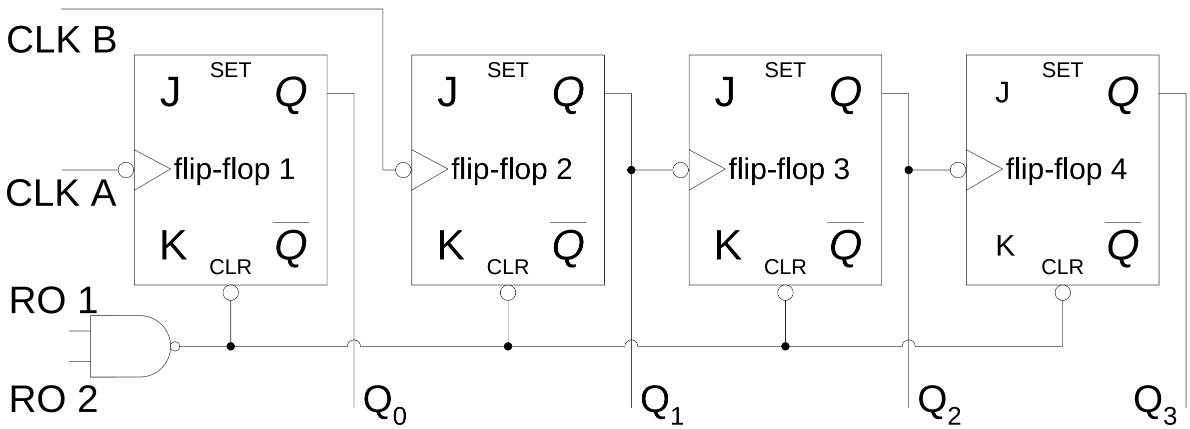
12. *Complete the timing diagram for the output* $\overline{CLK}$, ***Y*** *and* ***Q*** *by considering that you are applying given* ***CLK*** *and* ***D*** *as an input to the circuit given below where En is active high enable*

    Note*: Highlight the part of your timing diagram that indicates that's it's a Master SlaveFlip flop*

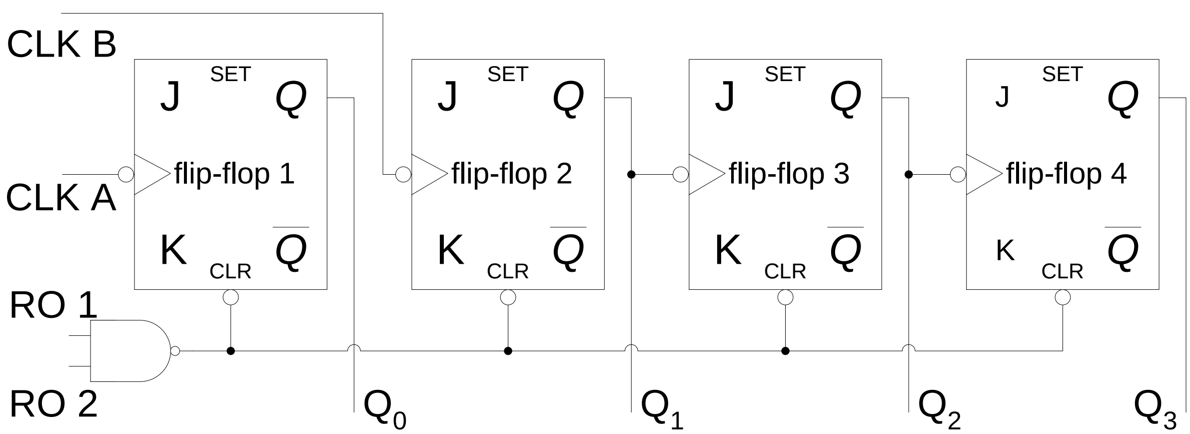




13. *Following is a **74LS93** is a **4-bit Asynchronous** Counter logic diagram? Convert it to **MOD(9)** Counter?*



14. Following is a **74LS93** is a **4-bit Asynchronous** Counter logic diagram? Convert it to **MOD(12)** Counter?



15. *Draw and label the circuit diagram of a* ***8-bit universal shift register*** *using eight* ***8 x 1 multiplexers*** *and eight* ***D flip flops****. The shift register should operate according to the following table.*

NOTE: *For reference see* ***Universal shift register*** *Section 6.2 Digital Design (Morris Mono, Global Edition)*

Mode Control			Register Operation
S2	S1	S0	
0	0	0	Reset to 0
0	0	1	Shift left
0	1	0	Set to 1
0	1	1	Complement the input
1	0	0	Shift right
1	0	1	Give 1's complement of the input
1	1	0	Parallel Load
1	1	1	Complement of DATA stored